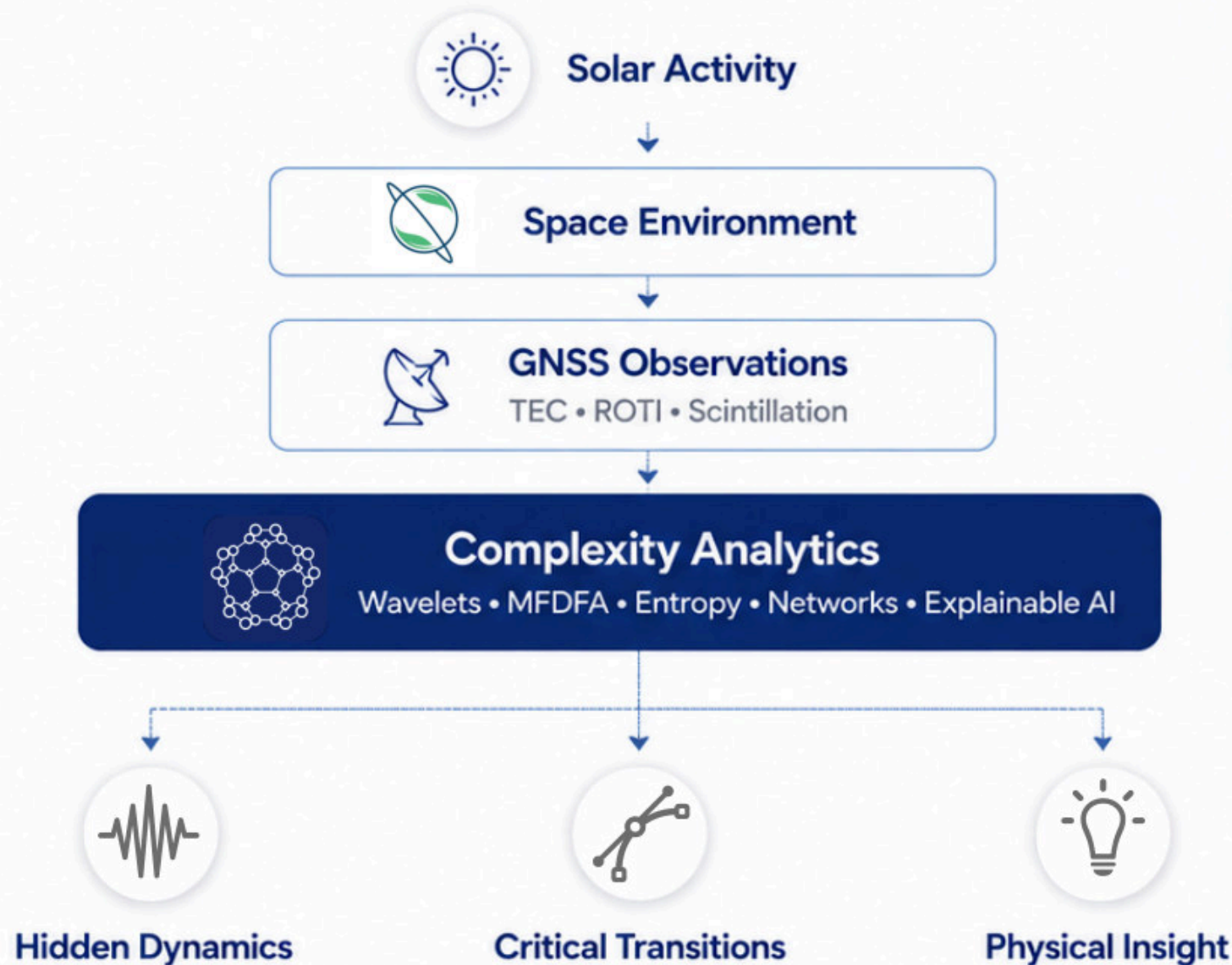


Complexity Analytics

for the Near-Earth Space Environment

A new framework for understanding ionospheric dynamics through nonlinear systems, complexity science and explainable AI.



From Traditional Analysis to Complexity Understanding

We move beyond prediction to uncover the hidden dynamics of the space environment.

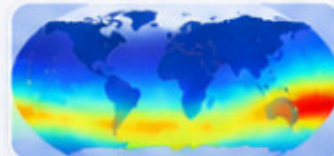
TODAY

What we typically do



TEC Maps

Global and regional representations



Statistical Analysis

Averages, trends, correlations



Forecasting

Empirical or physics-based models



ML Prediction

Black-box models for improved accuracy



Data-driven Approaches

Focus on correlation, not causation



We predict better, but we understand less.

NEXT GENERATION

What we aim to achieve



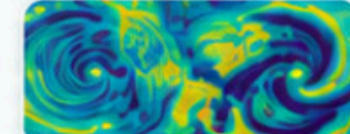
Complexity

Ionosphere as a complex nonlinear system



Emergence

Patterns emerge from interactions across scales



Criticality & Regime Shifts

Detect early signs of transitions and extreme events



Explainable AI

Interpretable models that link drivers to responses



New Physical Insight

From data-driven discovery to scientific understanding



We understand deeper, so we can anticipate better.



Goal: Reveal the hidden dynamics of the ionosphere

by integrating complexity science and explainable AI with large-scale observations.

Research Vision

Dr. Volkan Sarp

July 2026